Pregătire Metode numerice

Valori + vect. proprii

↳ există mai multe tehnici de deflație

1) Exemplu deflație → matrice simetrică

$$A = \begin{bmatrix} 4 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & -2 & 3 \end{bmatrix}$$

$\lambda_1 = 6$ → val. proprie dominantă

$$x_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

a) $B = (I_n - x_1 y^T) A$ a.î. $x_1^T y = 1$

$$y = \frac{1}{\|x_1\|^2} \cdot x_1$$ → Formule deflație

$$\Rightarrow y = \frac{1}{3} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

$$\Rightarrow y x_1^T = \frac{1}{3}\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\begin{bmatrix} 1 & -1 & 1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$B = \left(\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I_3} - \underbrace{\begin{bmatrix} 1/3 & -1/3 & 1/3 \\ -1/3 & 1/3 & -1/3 \\ 1/3 & -1/3 & 1/3 \end{bmatrix}}_{x_1 y^T} \right) \cdot A \Rightarrow$$

$$B = \begin{bmatrix} 2/3 & 1/3 & -1/3 \\ 1/3 & 2/3 & 1/3 \\ 1/3 & -1/3 & 1/3 \end{bmatrix} \begin{bmatrix} 4 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & -2 & 3 \end{bmatrix}$$

$$\Rightarrow \quad B = \frac{1}{3}\begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix}\begin{bmatrix} 4 & -1 & 1 \\ -1 & 3 & 2 \\ 1 & -2 & 3 \end{bmatrix} =$$

$$= \frac{1}{3}\begin{bmatrix} 6 & 3 & 1 \\ 3 & 3 & 8 \\ -3 & 0 & 7 \end{bmatrix}$$

$\lambda(B) = \{\lambda_2, \lambda_3, 0\}$

$x(B) = \{z_2, z_3, x_1\}$

$z_i = x_i - x_1 y^T x_i$

$z_2 = x_2 - x_1 \cdot y^T x_2$

$z_3 = z_3 - x_1 y^T x_3$

b) $A = \begin{bmatrix} 4 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ $\quad \lambda(A) = \{4, 2, 1\}$

Alt exemplu

$x_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $z_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $z_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

2) Givens 3×3 / 2×2 $\rightarrow$ valori + vectori proprii

$$G = \begin{bmatrix} c & s & 0 \\ -s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$c^2 + s^2 = 1$

$$\begin{vmatrix} \lambda - c & -s & 0 \\ s & \lambda - c & 0 \\ 0 & 0 & \lambda - 1 \end{vmatrix} = 0$$

De făcut calculele și de aflat valorile + vectorii proprii (ca la Algaed)

Valori proprii $\Rightarrow$ $\lambda_1 = 1$

$$\lambda_{2,3} = \frac{2c \pm \sqrt{4c^2 - 4}}{2} = c \pm is$$

Pt. matrice Givens de ord. 2 $\Rightarrow \lambda_{1,2} = c \pm is$

$$\begin{cases} \lambda_{1,2} = c \pm is \\ \lambda_{3,\dots,n} = 1 \end{cases} \rightarrow$$ toate valorile proprii au modulul 1 !

Metoda puterii inverse (MPI)

$$A = \begin{bmatrix} -4 & 14 & 0 \\ -5 & 13 & 0 \\ -1 & 0 & 2 \end{bmatrix} \qquad x^{(0)} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

μ = aproximație inițială = $\dfrac{x^{(0)T} A x^{(0)}}{x^{(0)T} \cdot x^{(0)}}$

calcule:

$$\begin{cases} x^{(0)T} \cdot x^{(0)} = 3 \\ x^{(0)T} \cdot A \cdot x^{(0)} = 19 \end{cases}$$

$$\Rightarrow \mu = \frac{19}{3}$$

$$A - \mu I = \begin{bmatrix} -4 - \frac{19}{3} & 14 & 0 \\ -5 & 13 - \frac{19}{3} & 0 \\ -1 & 0 & 2 - \frac{19}{3} \end{bmatrix} =$$

$$\begin{bmatrix} -\frac{31}{3} & 14 & 0 \\ -5 & \frac{20}{3} & 0 \\ -1 & 0 & -\frac{13}{3} \end{bmatrix}$$

1. $(A-\mu I)y^{(1)} = x^{(0)} \Rightarrow y^{(1)} = \dots$

$$y^{(1)} = \frac{y^{(1)}}{\|y^{(1)}\|}$$

$$\lambda^{(1)} = y^{(1)T} \cdot A \cdot y^{(1)}$$

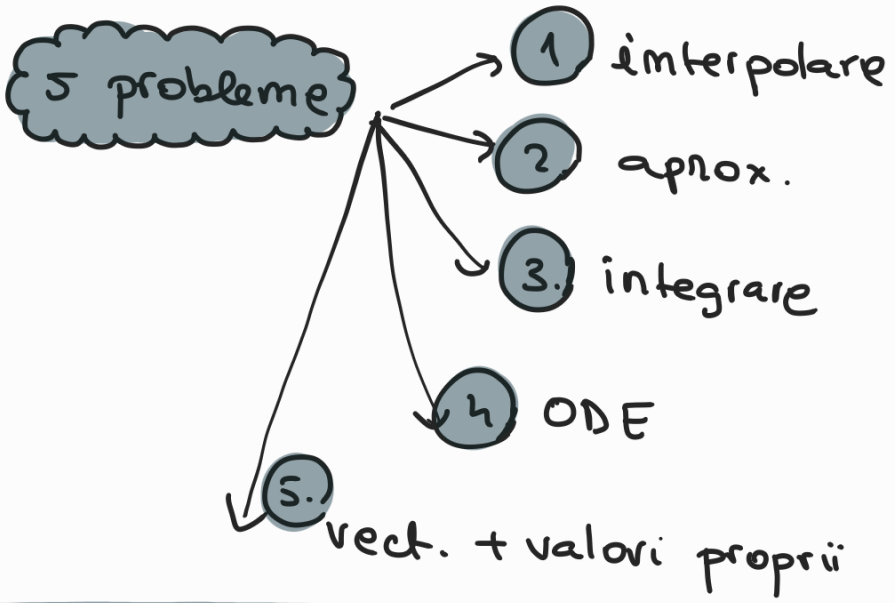
$$\mu = \lambda^{(1)}$$

2. $(A-\mu I)\cdot y^{(2)} = y^{(1)}$

$\Rightarrow y^{(2)} = \frac{y^{(2)}}{\|y^{(2)}\|}$ → formule pt. următorii pași

Examen

Test



2h/24 de minute pe întrebare

311CC_Florea_DARIA.pdf

Interpolare

tensionat $f' = [2 \; ? \; -1]$

1. Spline cubic natural / pt.

$x = [0 \quad 1 \quad 2]$ si $f = [-1 \quad 0 \quad 2]$

x	0	(1)	2
f	-1	0	2

$$S = \begin{cases} S_0(x) = a_0 + b_0 x + c_0 x^2 + d_0 x^3 \\ S_1(x) = a_1 + b_1(x-1) + c_1(x-1)^2 + d_1(x-1)^3 \end{cases}$$

→ de scris corect

condiții de interpolare

$S_0(0) = 1$ → $\boxed{a_0 = -1}$

$S_0(1) = 0$ → $a_0 + b_0 + c_0 + d_0 = 0$

$S_1(1) = 0$ → $\boxed{a_1 = 0}$

$S_1(2) = 2$ → $a_1 + b_1 + c_1 + d_1 = 2$

racordare

$S_0'(1) = S_1'(1)$ → $b_0 + 2c_0 + 3d_0 - b_1 = 0$

$S_0''(1) = S_1''(1)$ → $2c_0 + 6d_0 - 2c_1 = 0$

condiții la capete

$S_0''(0) = 0$ → $2c_0 = 0 \Rightarrow \boxed{c_0 = 0}$

$S_1''(2) = 0$ → $2c_1 + 6d_1 = 0$

$$\begin{cases} b_0 + d_0 = 1 \\ b_1 + c_1 + d_1 = 2 \\ b_0 + 3d_0 - b_1 = 0 \\ 3d_0 - c_1 = 0 \Rightarrow c_1 = 3d_0 \end{cases} + \quad c_1 + 3d_1 = 0$$

Calcule ---

$$\begin{cases} b_0 + d_0 = 1 \\ b_1 + 2d_0 = 2 \\ b_0 + 3d_0 - b_1 = 0 \end{cases} \qquad b_1 = b_0 + 3d_0$$

$$\left| \begin{array}{l} b_0 + d_0 = 1 \\ b_0 + 5d_0 = 2 \\ \hline 4d_0 - 1 = 0 \end{array} \right.$$

$$d_0 = 1/4$$

$$\begin{cases} d_1 = -\frac{1}{4} \\ c_1 = \frac{3}{4} \\ b_0 = \frac{3}{4} \\ b_1 = \frac{1}{2} \end{cases}$$

Suma coef. funcției Spline:

$$\underbrace{a_0 + b_0 + c_0 + d_0}_{\| \; 0} + a_1 + b_1 + c_1 + d_1 = 2$$

Lagrange → polinomul are gradul $n-1$

x	0	1	2
f	-1	0	2

$$P_2(x) = \sum_{i=1}^{3} f_i \prod_{\substack{j=1 \\ j \neq i}}^{3} \frac{x - x_j}{x_i - x_j} =$$

→ polinomul Lagrange

de sus bine ←

$$= (-1) \cdot \frac{(x-1)(x-2)}{(0-1)(0-2)} + 0 \cdot x - + \cancel{2} \cdot \frac{x(x-1)}{\cancel{(2-0)}(2-1)} =$$

$$= - \frac{(x-1)(x-2)}{2} + x(x-1) =$$

$$= \quad x^2 - x - \frac{x^2}{2} + \frac{3x}{2} - 1 = \frac{x^2}{2} + \frac{x}{2} - 1$$

$$P(1{,}5) = \frac{x}{2}(x+1) - 1 = \frac{1{,}5}{2} \cdot 2{,}5 - 1 = 0{,}875$$

Newton

x	F_0	F_1	F_2
0	−1	$\frac{0-(-1)}{1-0} = 1$	$\frac{2-(+1)}{2-0} = \frac{1}{2}$
1	0	$\frac{2-0}{2-1} = 2$	
2	2		

Atentie!

$$F_2[x_1, x_2, x_3] = \frac{F_1[x_2, x_3] - F_1[x_1, 2]}{x_3 - x_1}$$

$$P_2(x) = -1 + 1(x-0) + \frac{1}{2}(x-0)(x-1) =$$

$$= \frac{x^2}{2} - \frac{x}{2} + x - 1 = \frac{x^2}{2} + \frac{x}{2} - 1$$

$$P_2(0{,}5) = -0{,}625$$

$$P_2^{(L)}(1{,}5) + P_2^{(N)}(0{,}5) = 0{,}125$$

! Maxim 4 puncte pt. Lagrange

Aproximări

discretă

a)

x	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1
f	2	1	0	1	2

$g_2^*(x) = ?, \quad \omega(x) = 1$

$1 \quad x \quad x^2$

$$g_2^*(x) = c_0 + c_1 x + c_2 x^2$$

$$\begin{cases} c_0 \langle 1,1\rangle + c_1 \langle 1, x\rangle + c_2 \langle 1, x^2\rangle = \langle 1, f\rangle \\ c_0 \langle x,1\rangle + c_1 \langle x, x\rangle + c_2 \langle x, x^2\rangle = \langle x, f\rangle \\ c_0 \langle x^2,1\rangle + c_1 \langle x^2, x\rangle + c_2 \langle x^2, x^2\rangle = \langle x^2, f\rangle \end{cases}$$

$$\langle 1,1\rangle = \sum_{i=1}^{5} 1 = 5$$

$$\langle 1, x\rangle = \langle x, 1\rangle = \sum_{i=1}^{5} x_i = 0$$

$$\langle 1, x^2\rangle = \langle x^2, 1\rangle = \sum_{i=1}^{5} x_i^2 = 1 + \frac{1}{4} + 0 + \frac{1}{4} + 1 = \frac{5}{2}$$

$$= \langle x, x\rangle$$

$$\langle x, x^2\rangle = \langle x^2, x\rangle = \sum_{i=1}^{5} x_i^3 = 0$$

$$\langle x^2, x^2\rangle = \sum_{i=1}^{5} x_i^4 = 1 + \frac{1}{16} + 0 + \frac{1}{16} + 1 = \frac{34}{16} = \frac{17}{8}$$

$$\langle 1, f\rangle = \sum_{i=1}^{5} f_i = 2+1+0+1+2 = 6$$

$$\langle x, f\rangle = \sum_{i=1}^{5} x_i \cdot f_i = 0$$

$$\langle x^2, f\rangle = \sum_{i=1}^{5} f_i x_i^2 = 2 + \frac{1}{4} + 0 + \frac{1}{4} + 2$$

$$= \frac{9}{2} \Rightarrow \text{sistemul devine:}$$

$$\begin{cases} 5c_0 + \frac{5}{2}c_2 = 6 \\ \frac{5}{2}c_1 = 0 \\ \frac{5}{2}c_0 + \frac{17}{8}c_2 = \frac{9}{2} \end{cases} \Rightarrow \boxed{c_1 = 0}$$

$$\begin{cases} 5c_0 + \frac{5}{2}c_2 = 6 \\ 5c_0 + \frac{17}{4}c_2 = 9 \end{cases} \Rightarrow$$

$$\left(\frac{17}{4} - \frac{5}{2}\right) c_2 = 3 \iff \frac{17-10}{4} \cdot c_2 = 3 \iff$$

$$\frac{7}{4} \cdot c_2 = 3 \Rightarrow \boxed{c_2 = \frac{12}{7}} \Rightarrow$$

$$c_0 = \frac{1}{5}\left(6 - \frac{5}{2}c_2\right) = \frac{1}{5}\left(6 - \frac{30}{7}\right) =$$

$$= \frac{1}{5} \cdot \frac{12}{7} = \boxed{\frac{12}{35}} \Rightarrow$$

$$g_2^*(x) = \frac{12}{7}x^2 + \frac{12}{35}$$

$$e(0) = g_2^*(0) - f(0) = \frac{12}{35} = 0{,}34$$

b) $f:[-1,1] \to \mathbb{R}$, $f(x)=|x|$ continuă

$w(x)=\frac{1}{\sqrt{1-x^2}}$ $\quad w(x)=1$ → de calculat !

$$\langle T_0, T_0 \rangle = \pi$$

$$\langle T_1, T_1 \rangle = \frac{\pi}{2}$$

$$\langle T_2, T_2 \rangle = \frac{\pi}{2}$$

$$\langle T_0, f \rangle = \int_{-1}^{1} \frac{1 \cdot |x| \cdot \sqrt{1-x^2}}{\sqrt{1-x^2}}\, dx =$$

$$= \int_{-1}^{1} |x|\, dx = 2\int_{0}^{1} x\, dx = 1$$

$$\langle T_1, f \rangle = \int_{-1}^{1} x|x|\, dx = 0$$

$$\langle T_2, f \rangle = \int_{-1}^{1} (2x^2-1)\cdot|x|\, dx = 2\int_{0}^{1} (2x^3-x)\, dx$$

$$= 2\cdot\left(\frac{x^4}{2}-\frac{x^2}{2}\right)\Big|_0^1 = 0$$

$$g_2^*(x) = \frac{\pi}{\sqrt{2}} \cdot T_1(x)$$

Sist. normal

$$\begin{cases} c_0 \cdot \pi = 1 \\ c_1 \cdot \frac{\pi}{2} = 0 \\ c_2 \cdot \frac{\pi}{2} = 0 \end{cases} \Rightarrow g_2^*(x) = \frac{\pi}{\sqrt{2}}$$

Integrare numerica

- trapeze
- Simpson
- quadratura gaussiana

$$\int_{-1}^{1} f(x)\,dx = a_0 f(-1) + a_1 f(1) + b_0 f(x_1) + b_1 f(x_2)$$

$$\Pi_4 = \underbrace{(x-1)(x+1)}_{(x^2-1)}(x-x_1)(x-x_2)$$

$$(x^2 - Sx + P)$$

$$\begin{cases} \int_{-1}^{1} (x^2-1)(x^2-Sx+P)\,dx = 0 \\ \int_{-1}^{1} (x^2-1)(x^2-Sx+P)\cdot x\,dx = 0 \end{cases} \Rightarrow S, P = ?$$

$$\Rightarrow x_1, x_2 = ?$$

$f = 1 \qquad a_0 + a_1 + b_1 + b_0 = 2$

$f = x \qquad -a_0 + a_1 + b_0 x_1 + b_1 x_2 = 0$

$f = x^2 \qquad a_0 + a_1 + b_1 x_1^2 + b_2 x^2 = \frac{2}{3}$

$f = x^3$

Ecuații diferențiale

ecuație oscilație amortizată

$$\begin{cases} y'' + 2\delta y' + \omega^2 y = 0 \\ y(0) = 0 \\ v(0) = 1 \end{cases} \qquad y = y(t)$$

Ex. 1

a) Notăm $y' = v$ → viteza

$$v' + 2\delta v + \omega^2 y = 0$$

$$\begin{cases} y' = v \\ v' = -2\delta v + \omega^2 y \\ y(0) = 0 \\ v(0) = 1 \end{cases}$$

b) Pt. $\delta = -\frac{1}{2}$, $\omega = 1$ scrieți o metodă

RK2, $\alpha = 1$

$$\begin{cases} y' = v \\ v' = v - y \\ y(0) = 0 \\ v(0) = 1 \end{cases}$$

$Y_i = y(t_i)$

$V_i = v(t_i)$

$f_1(v, y, t) = v$

$f_2(v, y, t) = v - y$

$$\begin{cases} y' = f_1(v, y, t) \\ v' = f_2(v, y, t) \end{cases}$$ general

RK2)

$$\begin{cases} y_0 = 1 \\ v_0 = 1 \\ y_{i+1} = y_i + \dfrac{k_{y1} + k_{y2}}{2} \\ v_{i+1} = v_i + \dfrac{k_{v1} + k_{v2}}{2} \\ k_{y1} = f_1(v_i, y_i, t_i) = v_i \\ k_{v1} = f_2(v_i, y_i, t_i) = v_i - y_i \\ k_{y2} = f_1(v_i + k_{v1}, y_i + k_{y1}, t_i + h) = \\ \quad v_i + k_{v1} = 2v_i - y_i \\ k_{v2} = f_2(v_i + k_{v1}, y_i + k_{y1}, t_i + h) = \\ = v_i + k_{v1} - y_i - k_{yi} = v_i - y_i \end{cases}$$

$$y_{i+1} = y_i + \frac{v_i + 2v_i - y_i}{2} = \frac{1}{2}(3v_i - y_i)$$

$$v_{i+1} = v_i + \frac{v_i - y_i + v_i - 2y_i}{2} = 2v_i - \frac{3}{2}y_i =$$

$$= \frac{1}{2}(4v_i - 3y_i) \Rightarrow$$

$$\begin{cases} y_0 = 0 \\ v_0 = 1 \\ y_{i+1} = \frac{1}{2}(3v_i + y_i) \\ v_{i+1} = \frac{1}{2}(4v_i - 3y_i) \end{cases}$$

$$v_1 = \frac{1}{2} \cdot 4 = 2$$

$$y_1 = \frac{1}{2} \cdot 3 = \frac{3}{2}$$

$$v_2 = \frac{1}{2}\left(8 - \frac{9}{2}\right) =$$

$$= \frac{1}{2}\left(\frac{16-9}{2}\right) = \frac{1}{2} \cdot \frac{7}{2} = \frac{7}{4}$$

$$= 1{,}75$$

EX. 2 Definiți o metodă multi pas

$$y_{k+1} = y_k + h(a_0 f_k + a_1 f_{k-2})$$

+ condițiile inițiale $\begin{cases} y_0 = 0 \\ y_1 = 1 \end{cases}$ $f(x,y) = x^2 + y^2$

$h = 1$, $x_0 = 0$, $x_k = k$

$y = 1$ $\quad 0 = 0$

$y = x$ $\quad k+1 = k + a_0 + a_1 \Rightarrow a_0 + a_1 = 1$

$y = x^2$ $\quad (k+1)^2 = k^2 + 0 + 2(k-2)\, a_1 \Rightarrow$

$$\begin{cases} -4a_1 = 1 \\ a_0 + a_1 = 1 \end{cases} \Rightarrow \begin{cases} a_1 = -\frac{1}{3} \\ a_0 = \frac{5}{4} \end{cases} \Rightarrow$$

$$y_{k+1} = y_k + \frac{1}{4}(5f_k - f_{k-2})$$

Gradul maxim de valabilitate

$y = x^3$

$(k+1)^3 = k^3 + \frac{1}{4}(0.5 - 3(k-2)^2)$

$1 = -\frac{12}{4}$ (F) $\Rightarrow$ Dacă dădea adv. continuam cu x^4

gradul maxim de valabilitate este (2)

$$a_1^2 + a_0^2 = \frac{1}{16} + \frac{25}{16} = \frac{26}{16} = \frac{13}{8}$$